

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

LSTM Multi-Step Training Report

Report Generated: 2025-04-30 10:10:40

Training Start: 2025-04-17_16-14-17

Training End: 2025-04-17_17-32-02

Training Duration: 77.74 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.2331

MSE: 2.4797

RMSE: 1.5747

R2 Score: 0.5901

Multi-Step Forecast:

Forecast Steps: 10

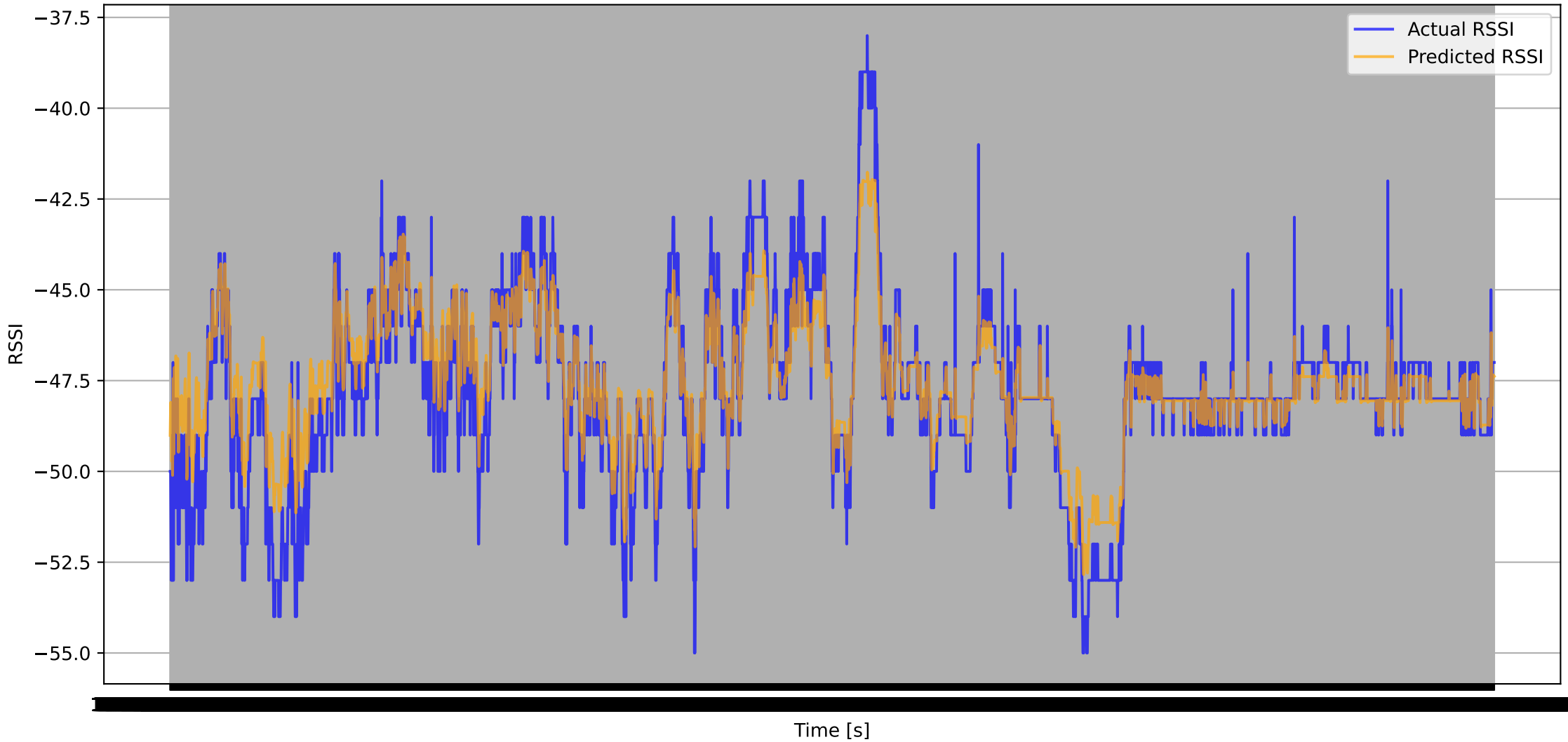
MAE: 0.0275

MSE: 0.0015

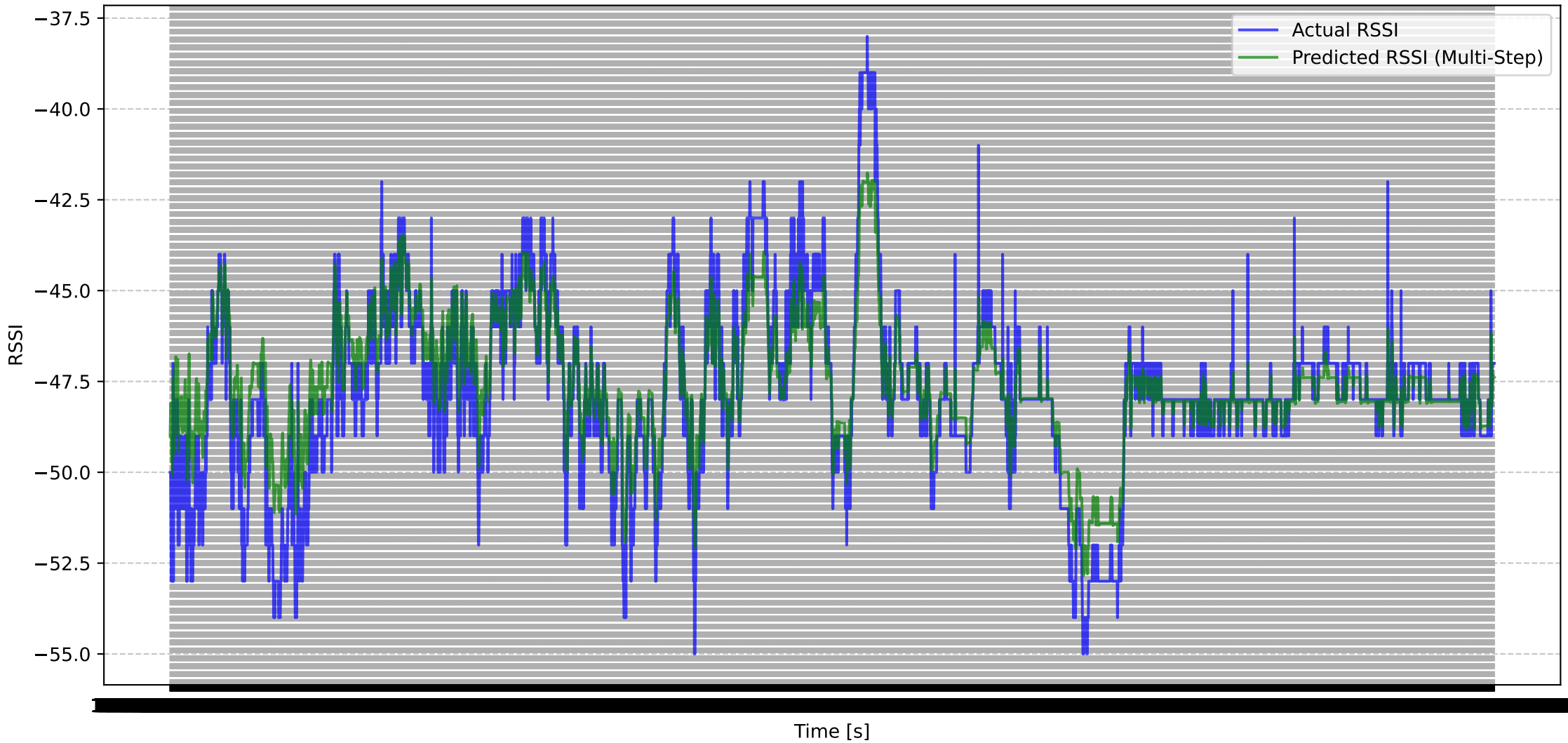
RMSE: 0.0383

R2 Score: 0.7963

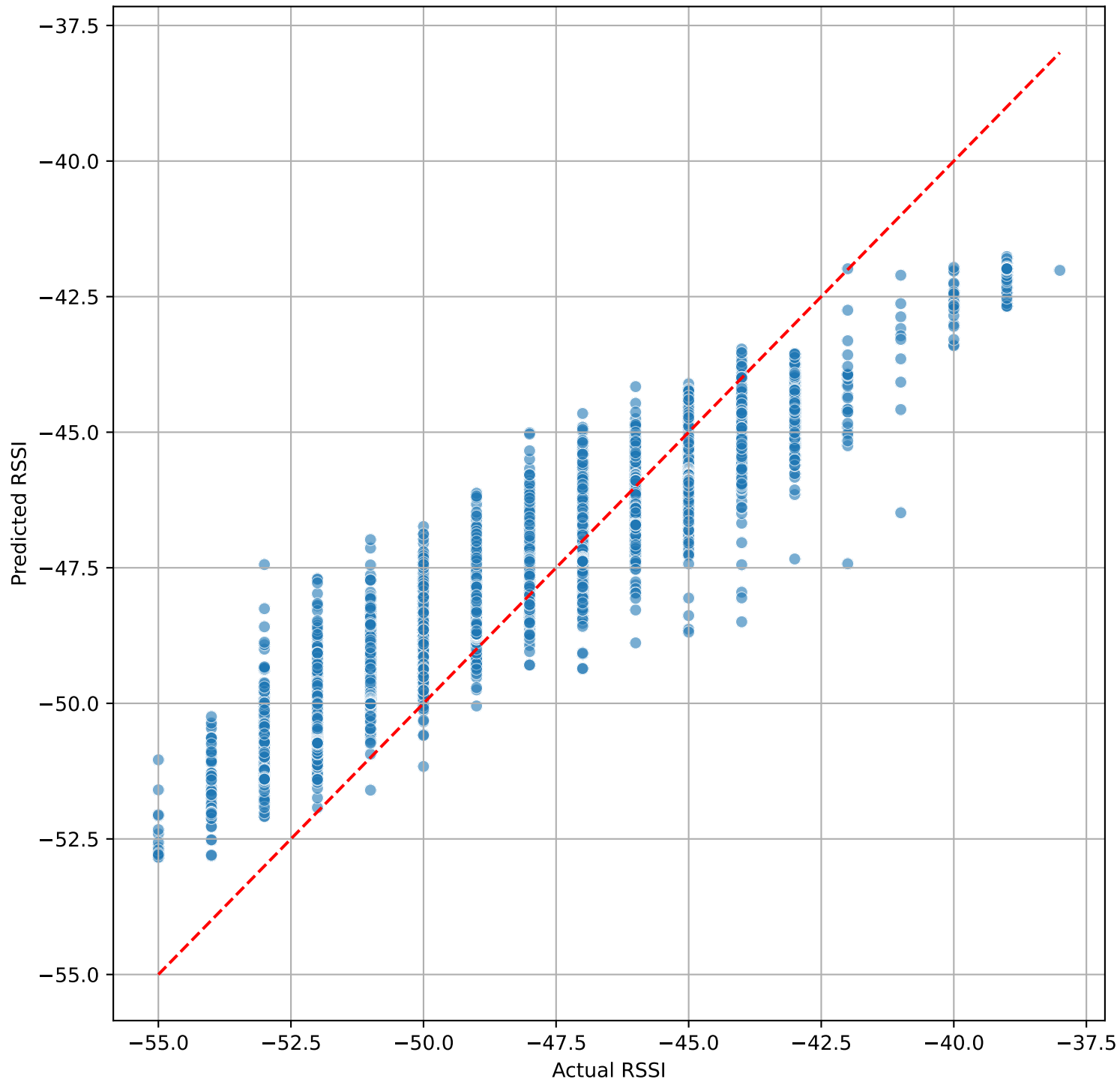
Actual vs Predicted RSSI (First Step)



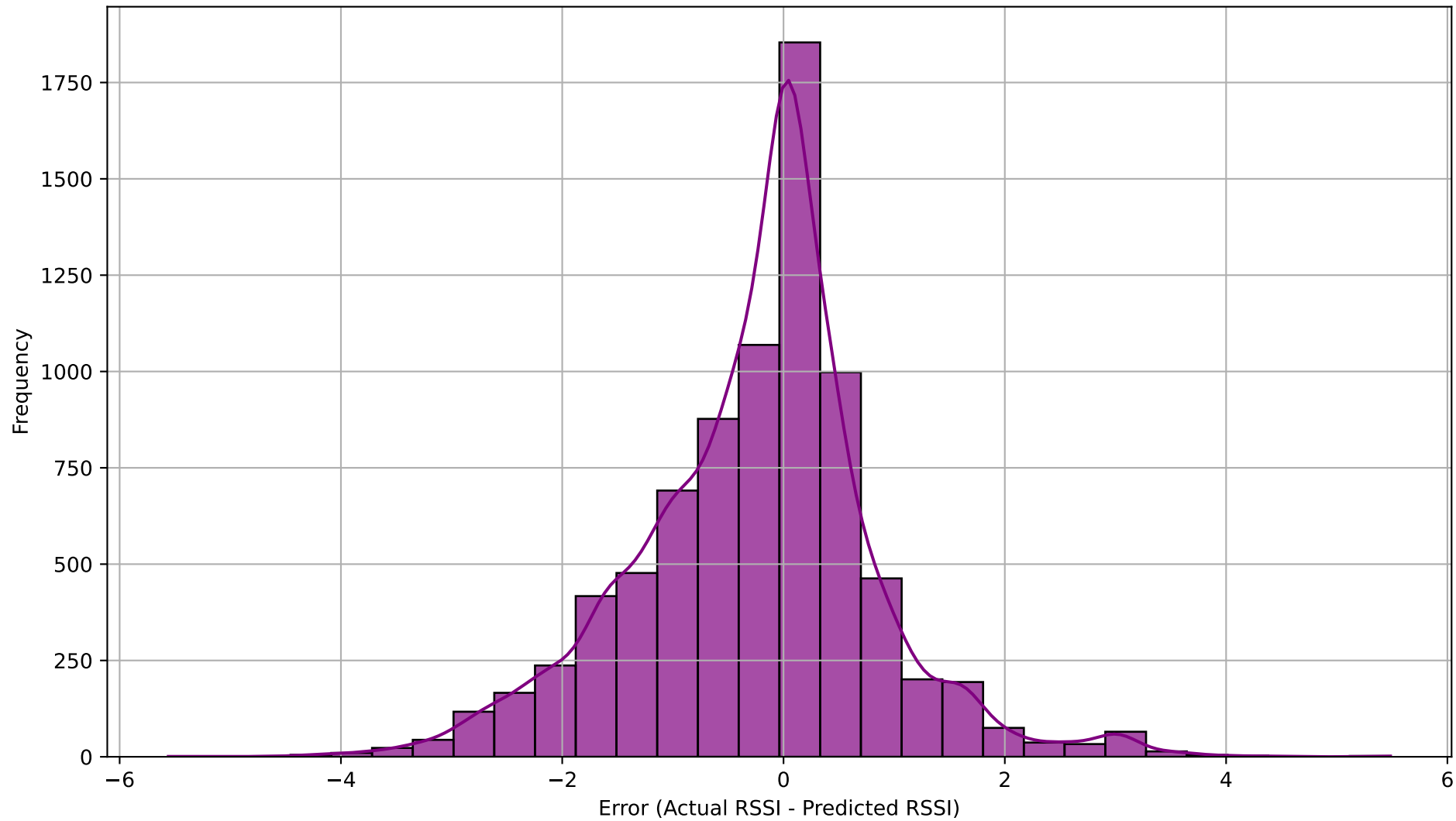
Actual vs Predicted RSSI (First Step vs Multi-Step)



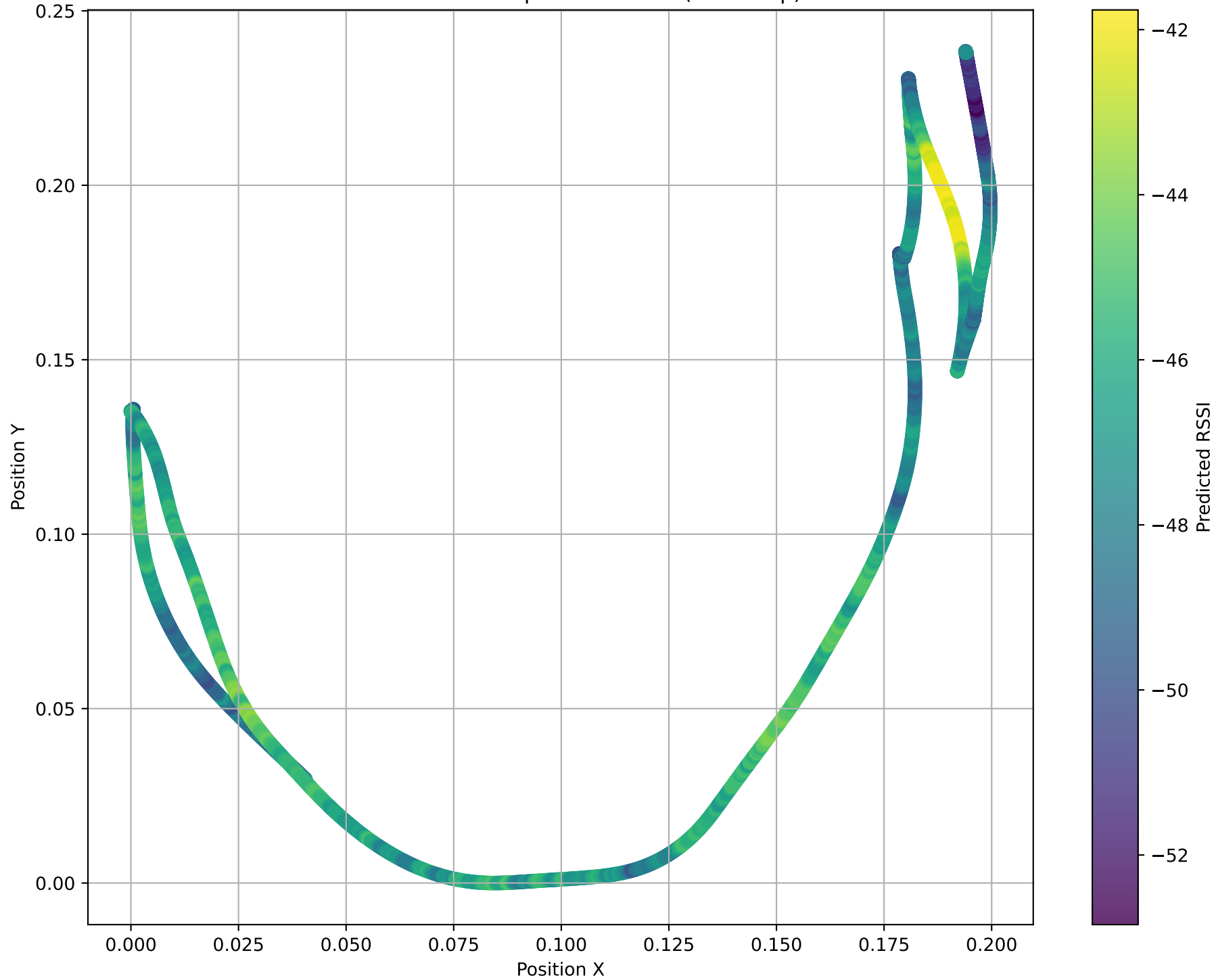
Actual vs Predicted RSSI (First Step)



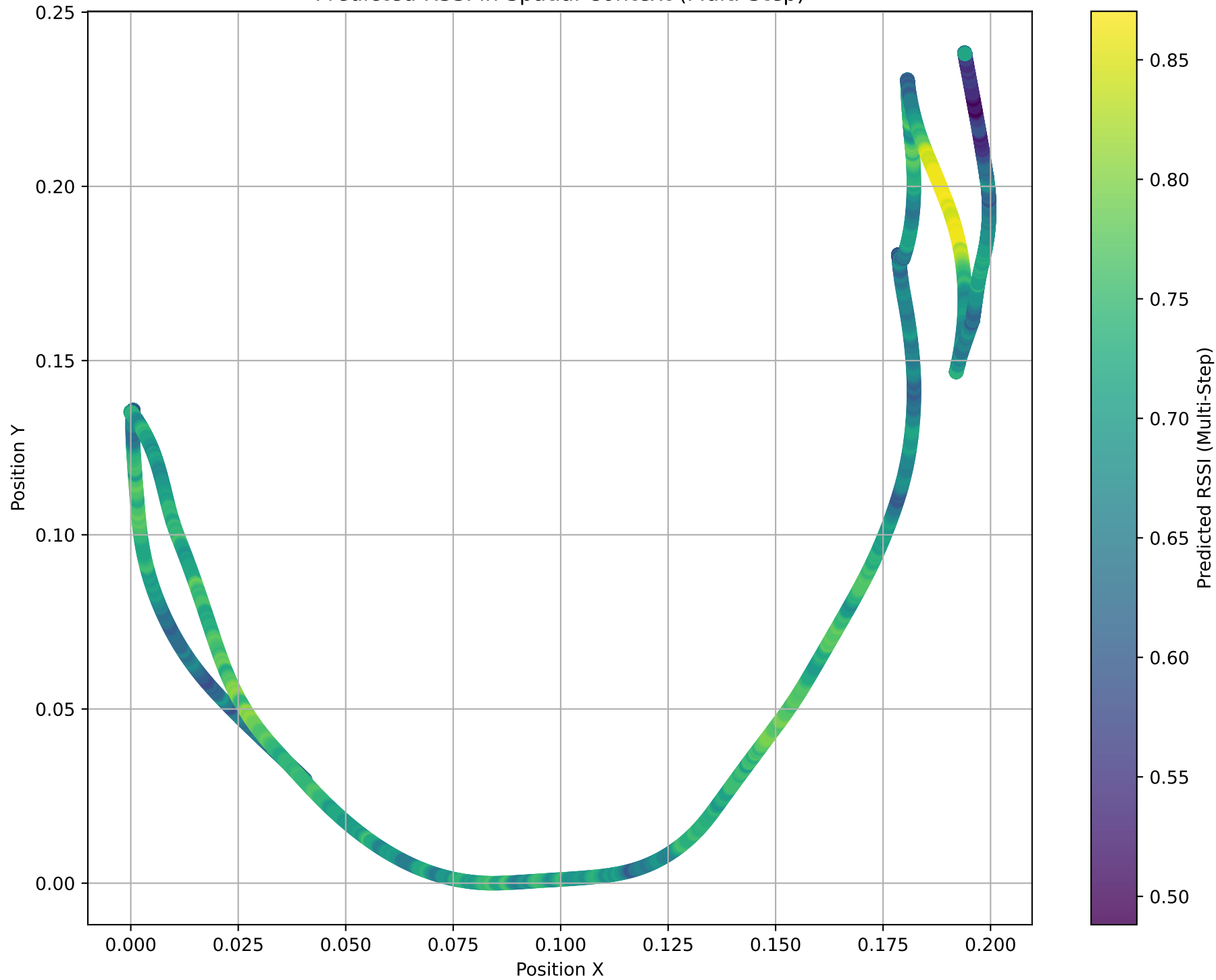
Distribution of Prediction Errors (First Step)



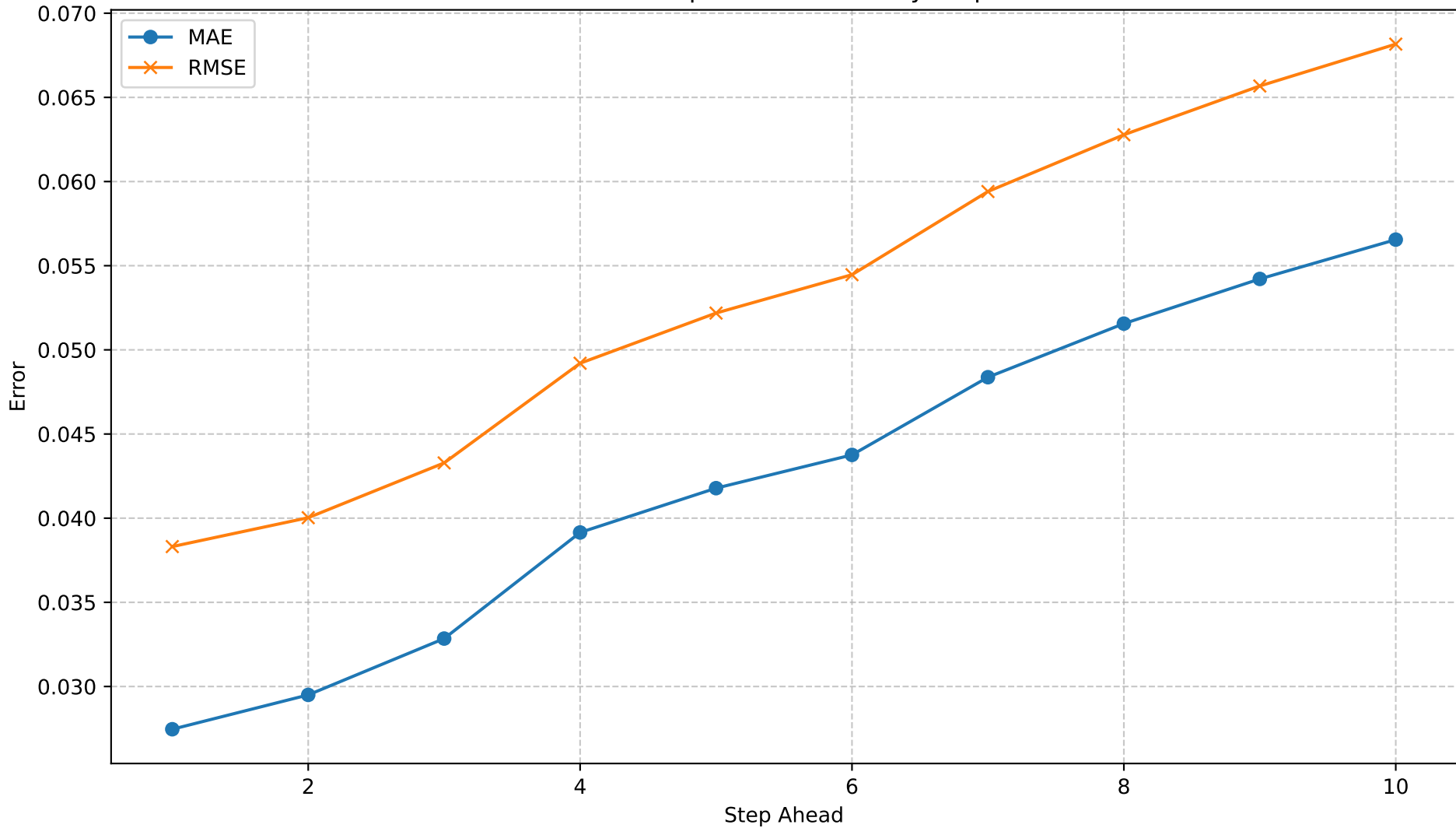
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step)



Multi-Step Forecast Error by Step



Resumo Automático:

- Forecast Steps configurados: 10
- Passos com MAE $\leq$ 2.0: 10 passo(s)
- MAE no último passo: 0.0565

[V] O modelo mantém boa acurácia durante quase toda a sequência de previsões.